

NOVA

Navigational Object and Voice Assistant

Software Requirements Specification (SRS) Version 1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional and non-functional requirements for NOVA (Navigational Object and Voice Assistant), a low-cost assistive vision mobile application designed for blind and visually impaired (BVI) users in Cameroon and broader Sub-Saharan Africa. NOVA targets Android devices running API level 26 (Android 8.0) and above, and is implemented using the Flutter framework.

This document is intended for use by the development team, academic supervisors, and future maintainers of the system. It serves as the authoritative reference for all design, implementation, testing, and evaluation decisions made throughout the project lifecycle.

1.2 Scope

NOVA is a smartphone-based assistive vision system that uses the Android device camera, on-device machine learning inference, cloud-based inference for heavy tasks, and text-to-speech output to help BVI users navigate their environment, read text, identify currency, recognise known faces, and obtain scene descriptions.

The system comprises four integrated components:

- A Flutter mobile application (Android) that captures camera frames, runs on-device ML inference, manages TTS output, and communicates with the backend.
- A FastAPI backend server that handles computationally expensive inference tasks (scene description, face recognition against server-side galleries) and serves updated ML models.
- A PostgreSQL relational database storing user profiles, enrolled faces, usage logs, and feedback.
- An ML pipeline (offline) producing quantized TensorFlow Lite models for obstacle detection and currency detection, deployed to the mobile app.

The following are explicitly out of scope: iOS support, clinical diagnostic features, BLE beacon infrastructure, production-scale deployment, and longitudinal field studies beyond the project evaluation period.

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
NOVA	Navigational Object and Voice Assistant: the application name
BVI	Blind and Visually Impaired, the primary user population
TTS	Text-to-Speech synthesis of audible speech from text
OCR	Optical Character Recognition conversion of image text to machine-readable text
CNN	Convolutional Neural Network deep learning architecture for image tasks
TFLite	TensorFlow Lite on-device ML inference runtime for Android
KD	Knowledge Distillation model compression technique

Term / Acronym	Definition
PTQ	Post-Training Quantization: quantization applied after training
QAT	Quantization-Aware Training: quantization applied during training
NFR	Non-Functional Requirement
FR	Functional Requirement
API	Application Programming Interface
REST	Representational State Transfer web API architectural style
JWT	JSON Web Token is a stateless authentication token
CFA	The Central African CFA franc is used in Cameroon
FPS	Frames Per Second
NPU	Neural Processing Unit, a dedicated ML accelerator on modern SoCs
IoT	Internet of Things
SDLC	Software Development Life Cycle
MoSCoW	Must / Should / Could / Won't requirement prioritisation framework

1.4 References

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- Rodriguez-Garavito et al. (2025). Computer vision-based obstacle detection. Multimodal Tech.

1.5 Overview

This SRS is organised as follows: Section 2 provides an overall description of the product, its user classes, constraints, and assumptions. Section 3 defines all functional requirements. Section 4 defines all non-functional requirements. Section 5 describes system interfaces. Section 6 covers data requirements. Section 7 addresses security requirements.

2. Overall Description

2.1 Product Perspective

NOVA is a new, standalone mobile application positioned in the ecosystem of smartphone-based assistive technology. Unlike existing solutions such as Microsoft Seeing AI (iOS-first), Envision AI (cloud-dependent, subscription model), and OrCam MyEye (USD 4,500 dedicated hardware), NOVA is designed from the ground up for:

- Low-end Android devices (quad-core CPU, 2 GB RAM, no dedicated NPU) dominate the Cameroonian market.
- Intermittent mobile data, all safety-critical features operate fully offline.
- The CFA franc currency zone and bilingual French/English Cameroonian context.
- A zero-cost-to-user model, the application is free with no subscription.

NOVA follows an IoT architecture: the Android smartphone acts as the edge endpoint for real-time inference; the FastAPI backend acts as the cloud service for heavy computation; and an operator dashboard (separate from this SRS) monitors system health and usage.

2.2 Product Functions (Summary)

NOVA delivers five core user-facing feature modules:

Module	ID	Description	Inference Location
Obstacle Detection	MOD-01	Real-time forward obstacle detection with directional TTS alerts	On-device (TFLite)
Text Reading (OCR)	MOD-02	Camera-based OCR with TTS read-aloud	On-device (ML Kit)
Scene Description	MOD-03	Natural-language scene description on user request	Cloud (FastAPI)
Currency Detection	MOD-04	CFA franc denomination identification	On-device (TFLite)
Face Recognition	MOD-05	Recognition and announcement of enrolled known contacts	Hybrid (TFLite detect + Cloud match)

2.3 User Classes and Characteristics

2.3.1 Primary User Blind or Visually Impaired Individual

The primary user is an adult (18–65) who is bilaterally blind or has severe visual impairment, owns or has access to an Android smartphone (API 26+), and communicates in French and/or English. This user interacts with NOVA exclusively through voice commands and audio output. They cannot rely on any visual interface element. Their primary needs are: safe navigation (obstacle avoidance), reading printed text in their environment, identifying banknotes, recognising familiar people, and understanding scenes they cannot perceive.

2.3.2 Secondary User Caregiver or Support Person

A caregiver or family member may assist with initial app setup, face enrolment, and emergency contact configuration. This user may have partial or full vision and can interact with visual interface elements during setup. After setup, this user is not involved in routine operation.

2.3.3 Operator / Developer

A developer or project team member accesses the backend operator dashboard to monitor usage logs, model performance, and aggregate feedback. This user interacts with the system entirely through the web dashboard and has no special mobile interface.

2.4 Operating Environment

Component	Specification
Mobile OS	Android 8.0 (API 26) through Android 15 (API 35)
Reference a low-end device	Quad-core 1.4 GHz, 2 GB RAM, 16 GB storage, no NPU
Reference mid-range device	Octa-core 2.0 GHz, 4 GB RAM, 64 GB storage, optional NPU
Camera	Rear camera, minimum 8 MP, autofocus
Network	2G/3G/4G LTE must function with no connectivity for core features
Backend server	Linux VPS, Python 3.11+, FastAPI, PostgreSQL 15
TTS engine	Android TextToSpeech API (on-device, no network required)
ML runtime	TensorFlow Lite 2.14+, Google ML Kit (on-device)

2.5 Design and Implementation Constraints

- All safety-critical features (obstacle detection, OCR, and currency detection) MUST function with zero network connectivity.
- On-device quantized model size per feature module MUST NOT exceed 20 MB.
- On-device inference latency for obstacle detection MUST remain below 100 ms per frame.
- The application MUST NOT transmit raw camera frames to the server at any time unless the user explicitly triggers a cloud feature (scene description, cloud face matching).
- All user-facing interactions MUST be operable via TalkBack screen reader and voice commands; no critical function may require visual feedback.
- The application MUST support Android TalkBack out-of-the-box (ContentDescription on all interactive elements).
- Battery consumption during continuous active use MUST remain below 25% per hour on the reference low-end device.
- The app MUST be distributable as a standard APK without requiring Google Play Store access.

2.6 Assumptions and Dependencies

- Users have a functional Android device with a working rear camera and speaker.
- Enrolled face images are provided by the user or caregiver during setup; NOVA does not perform unsolicited face surveillance.

- The FastAPI backend is hosted on a server with a stable internet connection; only the mobile device operates under intermittent connectivity.
- The TensorFlow Lite runtime is bundled within the Flutter app (not downloaded at runtime).
- Google ML Kit for OCR is used for on-device text recognition; it operates offline once downloaded.
- CFA franc note images used for training the currency detection model are sourced from the Bank of Central African States (BEAC) official reference images plus field-collected samples.

3. Functional Requirements

3.1 Module MOD-01: Obstacle Detection

3.1.1 Overview

MOD-01 provides continuous, real-time detection of obstacles in the user's forward field of view using the device's rear camera. The module runs a quantized object detection model on-device and issues directional audio alerts through the device speaker (or connected audio output) when obstacles are detected within configurable distance thresholds. MOD-01 is the highest-priority safety feature and must function fully offline.

3.1.2 Functional Requirements

ID	Requirement	Priority	Rationale
FR-01-01	The system SHALL activate the rear camera and begin processing frames when MOD-01 is active.	Must	Core precondition for all detection.
FR-01-02	The system SHALL run a quantized MobileNetV3-SSD or YOLOv8-nano TFLite model at a minimum of 10 FPS on the reference low-end device.	Must	Below 10 FPS, alerts arrive too late to be useful.
FR-01-03	The system SHALL detect obstacles from the 80-class COCO dataset plus an extended set of pedestrian-environment obstacles (open manholes, motorbikes, market stalls, low-hanging objects).	Must	COCO covers most obstacles; extension covers the Cameroonian context.
FR-01-04	When an obstacle is detected within the near-zone threshold (configurable, default 1.5 m estimated), the system SHALL immediately emit a TTS alert naming the obstacle and its approximate direction (left, centre, right) relative to the camera frame.	Must	Directional information is critical for navigation response.
FR-01-05	When an obstacle is detected within the warning-zone threshold (configurable, default 3.0 m estimated), the system SHALL emit a lower-urgency audio tone followed by a TTS alert.	Should	Graduated alerts reduce panic and improve navigation.
FR-01-06	The system SHALL suppress repeated alerts for the same tracked obstacle for a configurable suppression period (default 2.0 seconds) to prevent alert fatigue.	Must	Without suppression, continuous alerts cause users to disable the feature.
FR-01-07	The system SHALL use bounding box width and object class to estimate approximate distance in the absence of a depth sensor, using class-specific reference sizes.	Should	Monocular depth estimation from size is the only viable approach without dedicated hardware.

ID	Requirement	Priority	Rationale
FR-01-08	The system SHALL allow the user to toggle MOD-01 on/off via a single voice command ('Start obstacle detection', 'Stop obstacle detection').	Must	Full voice control is required for BVI users.
FR-01-09	The system SHALL continue obstacle detection with the screen off (background processing) to conserve battery.	Should	Screen-off operation significantly extends battery life during navigation.
FR-01-10	The system SHALL log detection events (timestamp, class, confidence, approximate zone) locally, and sync to the backend when connectivity is available.	Could	Supports operator analytics and model improvement.

3.2 Module MOD-02: Text Reading (OCR)

3.2.1 Overview

MOD-02 enables the user to read any printed or handwritten text visible to the camera. The user triggers OCR via a voice command or a single hardware button press. The system captures a still frame, processes it through Google ML Kit's on-device OCR engine, and reads the result aloud via TTS. MOD-02 operates fully offline.

3.2.2 Functional Requirements

ID	Requirement	Priority	Rationale
FR-02-01	The system SHALL capture a high-resolution still frame from the rear camera when the user triggers OCR.	Must	Higher resolution improves OCR accuracy on small text.
FR-02-02	The system SHALL process the captured frame using Google ML Kit Text Recognition v2 (Latin and French script support).	Must	ML Kit provides offline, on-device OCR with no additional cost.
FR-02-03	The system SHALL read the recognised text aloud in its detected reading order (top-to-bottom, left-to-right) via TTS.	Must	Reading order matters for comprehension of signs, documents, and labels.
FR-02-04	The system SHALL detect the dominant language of the recognised text and set the TTS language accordingly (English or French).	Should	The Cameroonian environment is bilingual; incorrect TTS language renders output incomprehensible.
FR-02-05	If no text is detected in the captured frame, the system SHALL announce 'No text detected. Try moving the camera closer.'	Must	Informative failure messages are required for BVI users who cannot see the frame.
FR-02-06	The system SHALL allow the user to re-trigger OCR immediately or cancel the current reading via voice command.	Must	Users need to correct mispositioning without navigating menus.

ID	Requirement	Priority	Rationale
FR-02-07	The system SHALL allow the user to adjust TTS reading speed (slow, normal, fast) via a voice command or settings.	Should	Reading speed preferences vary; dense documents may require a slower speed.
FR-02-08	The system SHALL support a continuous document reading mode where successive captures are stitched to read multi-page or multi-panel content.	Could	Useful for reading letters, forms, and pamphlets in sequence.
FR-02-09	The system SHALL log OCR trigger events (timestamp, text length, language detected) for analytics, with no raw text stored on the backend.	Could	Preserves user privacy while enabling usage analytics.

3.3 Module MOD-03: Scene Description

3.3.1 Overview

MOD-03 generates a natural-language description of the scene visible to the camera. This feature is cloud-dependent when triggered; a compressed still frame is sent to the FastAPI backend, which passes it to a vision-language model (VLM) and returns a text description that NOVA reads aloud. MOD-03 is explicitly not a safety feature and is documented as cloud-dependent to the user.

3.3.2 Functional Requirements

ID	Requirement	Priority	Rationale
FR-03-01	When the user triggers scene description via voice command, the system SHALL check network connectivity before proceeding.	Must	Scene description cannot function offline; the user must be informed.
FR-03-02	If no connectivity is available, the system SHALL announce 'Scene description requires an internet connection. Please try again when connected.' and abort.	Must	Offline failure must be graceful and informative.
FR-03-03	If connectivity is available, the system SHALL capture a JPEG still frame, compress it to a maximum of 512 KB, and transmit it to the FastAPI /scene/describe endpoint over HTTPS.	Must	Compression reduces data usage on expensive mobile plans.
FR-03-04	The FastAPI backend SHALL pass the image to a vision-language model (default: BLIP-2 or a cloud-hosted LLM with vision) and return a concise scene description (maximum 80 words).	Must	Concise descriptions are easier to process via audio.
FR-03-05	NOVA SHALL read the returned description aloud via TTS within 5 seconds of the user's trigger on a 4G connection.	Should	5 seconds is the upper bound for a perceptually responsive action.

ID	Requirement	Priority	Rationale
FR-03-06	The system SHALL announce 'Describing the scene, please wait.' immediately after the request is sent, to confirm the action was received.	Must	BVI users need audio confirmation that a cloud request is in progress.
FR-03-07	If the server returns an error or times out after 8 seconds, the system SHALL announce 'Scene description is unavailable right now. Please try again later.'	Must	Graceful timeout handling prevents silent failures.
FR-03-08	No raw image frames captured for scene description SHALL be stored persistently on the backend server; processing is transient.	Must	Privacy requirement: camera data must not accumulate on the server.

3.4 Module MOD-04: Currency Detection

3.4.1 Overview

MOD-04 identifies the denomination of Central African CFA franc banknotes (Series issued by BEAC: 500, 1000, 2000, 5000, 10000 FCFA) presented to the camera. The module uses a dedicated quantized TFLite classification model trained on CFA franc note images. Detection is fully on-device.

3.4.2 Functional Requirements

ID	Requirement	Priority	Rationale
FR-04-01	The system SHALL activate MOD-04 via voice command ('Identify money', 'Check note').	Must	Voice-only activation is required for BVI users.
FR-04-02	The system SHALL capture a still frame and process it through a TFLite classification model trained on all current BEAC CFA franc denominations (500, 1000, 2000, 5000, 10000 FCFA).	Must	All current denominations must be supported.
FR-04-03	The system SHALL announce the detected denomination (e.g., 'One thousand francs CFA') when confidence exceeds a threshold of 0.85.	Must	A high confidence threshold prevents false announcements that could mislead the user.
FR-04-04	If confidence is below 0.85, the system SHALL announce 'Could not identify the note clearly. Please try again with better lighting or closer to the camera.'	Must	Graceful low-confidence handling prevents silent errors.
FR-04-05	The system SHALL support both sides (obverse and reverse) of each denomination.	Should	Users may hold notes in either orientation; the model must be robust.
FR-04-06	The system SHALL function in typical indoor lighting conditions (200–800 lux); a torch/flashlight activation reminder SHALL be issued if the frame is detected as underexposed.	Should	Poor lighting is a common issue in Cameroonian indoor environments.

ID	Requirement	Priority	Rationale
FR-04-07	Detection SHALL complete within 500 ms of frame capture on the reference low-end device.	Must	Currency identification is a user-triggered, time-sensitive interaction.

3.5 Module MOD-05: Face Recognition

3.5.1 Overview

MOD-05 allows the user to enrol known contacts (family members, colleagues, caregivers) by providing photos during setup. When the user triggers face recognition, the front or rear camera scans the scene, detected faces are matched against the enrolled gallery, and recognised individuals are announced by name via TTS. Detection occurs on-device; matching against the enrolled gallery may be performed on-device (if the gallery is small, ≤ 20 faces) or via a backend API call (for larger galleries).

3.5.2 Functional Requirements

ID	Requirement	Priority	Rationale
FR-05-01	The system SHALL allow a sighted caregiver or the user to enrol contacts during setup by capturing 3–5 photos of each person and providing a spoken name label.	Must	Enrolment is the prerequisite for all face recognition functionality.
FR-05-02	The system SHALL detect faces in the camera frame using an on-device TFLite face detection model (BlazeFace or equivalent) at a minimum of 10 FPS.	Must	On-device detection ensures faces are found even offline.
FR-05-03	For galleries of 20 or fewer enrolled contacts, the system SHALL perform face embedding comparison entirely on-device using a quantized MobileFaceNet model.	Must	Fully offline for small galleries.
FR-05-04	For galleries exceeding 20 enrolled contacts, the system SHALL transmit detected face crops (not full frames) to the FastAPI /faces/match endpoint for server-side matching.	Should	Server-side matching scales to larger galleries without increasing device memory.
FR-05-05	When a match is found with cosine similarity above the threshold (default: 0.75), the system SHALL announce the contact's name via TTS.	Must	Threshold prevents false positive announcements.
FR-05-06	When no match is found, the system SHALL announce 'Unknown person detected' if a face was detected, or remain silent if no face is detected.	Must	Distinguish between a no-face and an unknown face.
FR-05-07	The system SHALL not perform continuous unsolicited face scanning; MOD-05 SHALL only activate on explicit user voice command.	Must	Privacy requirement: Passive surveillance is explicitly prohibited.

ID	Requirement	Priority	Rationale
FR-05-08	Face embedding vectors stored on the backend SHALL be encrypted at rest; raw enrolment images SHALL NOT be retained after embedding extraction.	Must	Biometric data privacy; embeddings are less sensitive than raw images.
FR-05-09	The user SHALL be able to delete any enrolled contact and their associated embeddings via voice command.	Must	Right to erasure for biometric data.

3.6 Cross-Cutting Functional Requirements

ID	Requirement	Priority
FR-06-01	The system SHALL provide a fully voice-navigable main menu with modules addressable by spoken command or TalkBack swipe.	Must
FR-06-02	The system SHALL support bilingual TTS output in English and French, selectable in settings.	Must
FR-06-03	The system SHALL provide haptic feedback (vibration) as a secondary alert channel for obstacle detection and face recognition, to support users in noisy environments.	Should
FR-06-04	The system SHALL allow the user to set an emergency contact and trigger location sharing via voice command ('Call for help', 'Send my location').	Should
FR-06-05	The system SHALL provide an onboarding tutorial narrated entirely via TTS at first launch, covering all five modules.	Must
FR-06-06	The system SHALL sync usage event logs (module activations, errors, user feedback ratings) to the backend when the network is available, using a queue that persists across app restarts.	Could
FR-06-07	The system SHALL support OTA model updates when a new quantized model is available on the backend, the app SHALL download and hot-swap it on next launch, with user notification via TTS.	Should
FR-06-08	The system SHALL allow the user to rate the usefulness of any interaction	Could

ID	Requirement	Priority
	(good/bad) via a voice command immediately after an interaction.	
FR-06-09	After downloading a model update via OTA, the system SHALL verify the SHA-256 checksum of the downloaded file against the value stored in the model registry. If verification fails, the system SHALL discard the downloaded file, retain the previously installed model, and announce "Model update failed. Using previous version." via TTS.	Must

4. Non-Functional Requirements

4.1 Performance Requirements

ID	Requirement	Target	Measurement Method
NFR-01	Obstacle detection frame rate (low-end device)	≥ 10 FPS	Stopwatch + Android Profiler during 60s walking session
NFR-02	Obstacle detection inference latency	< 100 ms per frame	TFLite inference time logging
NFR-03	OCR processing time (A4 document, on-device)	< 3 seconds	Measured from trigger to first TTS word
NFR-04	Currency detection latency	< 500 ms per trigger	Measured from trigger to TTS announcement
NFR-05	Scene description end-to-end latency (4G)	< 5 seconds	Measured from trigger to first TTS word
NFR-06	Face detection frame rate	≥ 10 FPS	Android Profiler
NFR-07	App cold start time	< 4 seconds	Measured from launch icon tap to main menu TTS
NFR-08	TTS announcement initiation delay	< 200 ms from trigger	Audio timestamp comparison

4.2 Model Size and Memory Requirements

ID	Requirement	Target
NFR-09	Quantized obstacle detection model size (TFLite INT8)	< 15 MB
NFR-10	Quantized currency detection model size (TFLite INT8)	< 10 MB
NFR-11	On-device face detection model size (BlazeFace)	< 5 MB
NFR-12	On-device face embedding model size (MobileFaceNet INT8)	< 12 MB
NFR-13	Total on-device ML model footprint	< 45 MB
NFR-14	App RAM usage during obstacle detection (foreground)	< 350 MB
NFR-15	App RAM usage during background detection (screen off)	< 200 MB

4.3 Battery and Energy Requirements

ID	Requirement	Target
NFR-16	Battery drain during continuous obstacle detection (screen off)	$< 25\%$ per hour on the reference device

ID	Requirement	Target
NFR-17	Battery drain during standby (app backgrounded, no active module)	< 3% per hour
NFR-18	Camera warm-up and model loading energy	Included in the per-hour measurement above

4.4 Reliability and Availability

ID	Requirement	Target
NFR-19	App crash rate during obstacle detection session	< 1 crash per 100 sessions
NFR-20	Graceful recovery from network disconnection	All offline features continue without a restart
NFR-21	Graceful recovery from camera access failure	TTS error message; no crash
NFR-22	Graceful recovery from TFLite model loading failure	TTS error message; alternative cloud fallback offered
NFR-23	Backend API availability target	>= 99% uptime (project evaluation period)
NFR-24	Offline queue persistence for usage logs	Logs persist across app restart and device reboot

4.5 Security Requirements

ID	Requirement
NFR-25	All backend API calls SHALL use HTTPS (TLS 1.2 or higher); plaintext HTTP is not permitted.
NFR-26	User authentication SHALL use JWT tokens with a 24-hour expiry and refresh token rotation.
NFR-27	Face embedding vectors SHALL be stored encrypted at rest on the backend (AES-256).
NFR-28	Raw camera frames SHALL NOT be transmitted to the backend except on explicit user request (MOD-03, MOD-05 cloud path); the backend SHALL NOT log image data.
NFR-29	API endpoints SHALL enforce rate limiting (60 requests/minute per authenticated user) to prevent abuse.
NFR-30	User passwords SHALL be hashed using bcrypt with a minimum cost factor of 12.
NFR-31	The mobile app SHALL NOT request permissions beyond camera, microphone, location (emergency feature), internet, and vibrate.

4.6 Usability and Accessibility Requirements

ID	Requirement
NFR-32	All interactive UI elements SHALL have a ContentDescription attribute for TalkBack compatibility.

ID	Requirement
NFR-33	No critical function SHALL require visual feedback; all confirmations, errors, and results SHALL be communicated via TTS.
NFR-34	Voice command recognition SHALL function with typical ambient noise levels (up to 65 dB, equivalent to a busy street).
NFR-35	TTS speech rate SHALL be configurable between 0.5x and 2.0x normal rate.
NFR-36	The onboarding tutorial SHALL be completable in under 5 minutes without sighted assistance.
NFR-37	The app SHALL support Android font scaling and display size settings without layout breakage (for low-vision users who use both visual and audio interfaces).

4.7 Maintainability and Portability

ID	Requirement
NFR-38	The Flutter codebase SHALL follow a feature-first folder structure with clearly separated domain, data, and presentation layers.
NFR-39	The FastAPI backend SHALL be containerised (Docker) for reproducible deployment.
NFR-40	TFLite models SHALL be versioned and served through the backend model registry; the mobile app SHALL support model hot-swapping without reinstallation.
NFR-41	All API endpoints SHALL be documented via FastAPI auto-generated OpenAPI (Swagger) documentation.
NFR-42	Unit test coverage for the FastAPI backend SHALL exceed 70% of business logic.

5. External Interface Requirements

5.1 User Interfaces

NOVA's mobile UI is designed around the principle of audio-first interaction. Visual elements are present for sighted caregivers and setup scenarios, but are never the sole interface for any function.

- Main Menu: Five large, high-contrast buttons (one per module) with icons and spoken labels. TalkBack-navigable. Voice shortcut overlay always active.
- Settings Screen: Voice-navigable list of configurable parameters (TTS speed, thresholds, language, emergency contact). All changes confirmed via TTS.
- Enrolment Screen (MOD-05): Camera viewfinder with audio guidance for centering the face; the sighted caregiver can verify the frame before capture.
- Status Overlay: A minimal always-visible strip showing active module and connectivity status, readable by TalkBack.

5.2 Hardware Interfaces

- Rear Camera: Accessed via Flutter camera plugin (android.hardware.camera2 API). Minimum 8 MP, autofocus required.
- Microphone: Accessed via Flutter speech_to_text plugin for voice command recognition.
- Speaker/Audio Output: Accessed via Android TextToSpeech API and audio_session plugin for routing to Bluetooth if connected.
- Vibration Motor: Accessed via Flutter vibration plugin for haptic alerts.
- GPS: Accessed via Flutter geolocator plugin for emergency location sharing (MOD emergency feature).

5.3 Software Interfaces

Interface	Direction	Protocol/API	Purpose
Google ML Kit Text Recognition v2	Local library	Dart FFI / plugin	On-device OCR
TensorFlow Lite	Local library	tf lite_flutter plugin	On-device CNN inference
Android TextToSpeech	Local OS API	Platform channel	Audio output
FastAPI Backend /scene/describe	Outbound	HTTPS REST POST	Cloud scene description
FastAPI Backend /faces/match	Outbound	HTTPS REST POST	Cloud face matching
FastAPI Backend /auth/*	Outbound	HTTPS REST POST	User authentication
FastAPI Backend /models/latest	Outbound	HTTPS REST GET	OTA model updates

Interface	Direction	Protocol/API	Purpose
FastAPI Backend /logs/sync	Outbound	HTTPS REST POST	Usage log sync
Android SMS Manager	Outbound	Platform channel	Emergency location SMS

5.4 Communication Interfaces

- HTTPS (TLS 1.2+): All client-server communication. Port 443.
- JSON: Request and response body format for all API endpoints.
- JWT Bearer tokens: Passed in the Authorization header for all authenticated endpoints.
- Offline queue: SQLite local database (drift ORM in Flutter) stores unsynced log events; flushed to the backend on the next available connection.

6. Data Requirements

6.1 Logical Data Model (Summary)

The system manages six primary data entities:

Entity	Key Attributes	Storage Location
User	user_id, email, password_hash, preferred_language, created_at	PostgreSQL (backend)
EnrolledFace	face_id, user_id, contact_name, embedding_vector (encrypted), created_at	PostgreSQL (backend)
UsageEvent	event_id, user_id, module_id, timestamp, outcome, confidence_score	SQLite (device) + PostgreSQL (synced)
UserFeedback	feedback_id, event_id, rating (positive/negative), timestamp	SQLite (device) + PostgreSQL (synced)
ModelRegistry	model_id, module_id, version, filename, checksum, uploaded_at, is_active	PostgreSQL (backend)
EmergencyContact	contact_id, user_id, name, phone_number, encrypted	PostgreSQL (backend)

6.2 Data Retention Policy

- Raw camera frames: Never persisted. Processed in memory; discarded after inference.
- Scene description images: Transmitted to backend for processing; not stored after response is generated.
- Face enrolment images: Used to generate embeddings; not stored after embedding extraction on the backend.
- Usage event logs: Retained on the backend for 12 months for analytics; anonymised after 12 months.
- Face embedding vectors: Retained until the user explicitly deletes the enrolled contact.
- JWT refresh tokens: Expire after 30 days; invalidated on explicit logout.

7. Verification and Validation Requirements

Requirement ID	Verification Method	Acceptance Criterion
FR-01-02	Automated benchmark on reference device	≥ 10 FPS sustained over a 60-second test
FR-01-06	Unit test: alert suppression logic	No repeat alert within 2.0 s for the same tracked object ID
FR-02-02	Functional test: 10 printed text samples	$\geq 90\%$ word accuracy on clear printed text
FR-03-05	End-to-end timing test on 4G	< 5 s from trigger to first TTS word in 90% of trials
FR-04-03	Functional test: 50 note images per denomination	$\geq 95\%$ accuracy when confidence > 0.85
FR-05-05	Functional test: enrolled a gallery of 10 contacts	$\geq 90\%$ recognition rate; $< 5\%$ false positive rate
NFR-01	Android Profiler, 60 s walking test	≥ 10 FPS on reference device
NFR-16	Battery drain test: 1 hour active detection	$< 25\%$ battery drop on reference device
NFR-25	Network traffic inspection (Charles Proxy)	Zero plaintext HTTP requests observed
NFR-32	TalkBack automated audit (Accessibility Scanner)	Zero critical accessibility errors